

Predictive Modeling

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BIG DATA

Predictive Modeling

- A powerful tool to turn data into action.
- It works because God made the universe predictable (and successful prediction rewarding)
- **Need for wisdom:** It can be used for great good and great harm

Power of Predictive Modeling

- **Medicine**: wearable monitor for seizures or falls, detect malaria from blood smears, find effective drug regimens from medical records
- **Drug Discovery**: predict the efficacy of a synthesis plan for a drug
- **Precision Agriculture**: predict effect of micro-climate on plant growth
- **Urban Planning**: forecast resource needs, extreme weather risks, ...
- **Government**: classify feedback from constituents
- **Retail**: predict items in a grocery order
- **Recommender systems**: Amazon, Netflix, YouTube, ...
- **User interfaces**: gesture typing,

and so much more...

The universe is predictable

- God created the world with actionable structure
 - We act rightly when our perceptions align with that structure
 - Every living thing does! All living things take in information, make inferences about the world from it, and use those inferences to inform actions.
 - We can discover that structure by learning to be less surprised by what we see (= predicting our perceptions)
- Perceptions are thus both accurate and fallable.

Need for wisdom

- Potential for great good
- But also great harm:
 - Lack of **fairness** in facial recognition, sentencing, lending, job applicant scoring, ...
 - Lack of **transparency** in how "Big Data" systems make conclusions
 - Lack of **privacy** as data is increasingly collected and aggregated
 - Amplification of extreme positions in social media, YouTube, etc.

Back down to earth...

Stating and refining the question

Six types of questions

1. **Descriptive:** summarize a characteristic of a set of data
2. **Exploratory:** analyze to see if there are patterns, trends, or relationships between variables (hypothesis generating)
3. **Inferential:** analyze patterns, trends, or relationships in representative data from a population
4. **Predictive:** make predictions for individuals or groups of individuals
5. **Causal:** whether changing one factor will change another factor, on average, in a population
6. **Mechanistic:** explore "how" as opposed to whether

Leek, Jeffery T., and Roger D. Peng. "What is the question?." Science 347.6228 (2015): 1314-1315.

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6. **Mechanistic:** how a diet high in fresh fruits and vegetables leads to a reduction in the severity of viral illnesses

Our focus: Prediction

- **Why** are we doing this?
 - Predictions enable decision-making
 - trying to predict helps us understand.
- **What** are we doing?
 - Predict something unknown from something known. Specifically: complete-the-table model
 - Mostly we'll *assume independent observations* (e.g., generalize across people)
 - Sometimes we'll *forecast* (e.g., predict how a time series will continue)
- **How**: We'll look at...
 - methods that consider similar examples (Nearest Neighbors)
 - methods that look at overall trends (linear/logistic regression)
 - more advanced methods, time permitting

Example: Home Sale Prices

From Ames, Iowa home sales, 2006-2010. (De Cock, 2011)

Lot_Area	Total_Bsmt_SF	Gr_Liv_Area	Garage_Cars	Sale_Price
31770	1080	1656	2	215000
11622	882	896	1	105000
14267	1329	1329	1	172000
11160	2110	2110	2	244000
13830	928	1629	2	189900

(2930 total rows)

- *Y: response variable (aka outcome, dependent variable):* **Sale_Price**
- *X: features (aka predictors, covariates, etc.):* everything else